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Cryptology Assignment 4

1. Decrypt the following ciphertext using Hill cipher.

i) CT : HLDTQIHJDXIWQCMALJ , $Key = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$

Inverse

$\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} = (3\times7) - (5\times2) = 21 - 10 = 11$, $11^{-1} \bmod 26 = 19$

$P = K^{-1} C \bmod 26$

$19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \bmod 26 = \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \bmod 26 = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$

$\begin{bmatrix} 3 & 14 \\ 9 & 15 \end{bmatrix} \begin{bmatrix} 7 \\ 11 \end{bmatrix} \bmod 26 = \begin{bmatrix} 3\times7 + 14\times11 \\ 9\times7 + 5\times11 \end{bmatrix} \bmod 26 = \begin{bmatrix} 175 \\ 118 \end{bmatrix} \bmod 26 = \begin{bmatrix} 19 \\ 14 \end{bmatrix}$

$\begin{bmatrix} 3 & 14 \\ 9 & 15 \end{bmatrix} \begin{bmatrix} 3 \\ 19 \end{bmatrix} \bmod 26 = \begin{bmatrix} 275 \\ 322 \end{bmatrix} \bmod 26 = \begin{bmatrix} 15 \\ 12 \end{bmatrix}$

$\begin{bmatrix} 3 & 14 \\ 9 & 15 \end{bmatrix} \begin{bmatrix} 16 \\ 8 \end{bmatrix} \bmod 26 = \begin{bmatrix} 160 \\ 184 \end{bmatrix} \bmod 26 = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$

$\begin{bmatrix} 3 & 14 \\ 9 & 15 \end{bmatrix} \begin{bmatrix} 7 \\ 9 \end{bmatrix} \bmod 26 = \begin{bmatrix} 147 \\ 198 \end{bmatrix} \bmod 26 = \begin{bmatrix} 17 \\ 4 \end{bmatrix}$

$\begin{bmatrix} 3 & 14 \\ 9 & 15 \end{bmatrix} \begin{bmatrix} 3 \\ 23 \end{bmatrix} \bmod 26 = \begin{bmatrix} 331 \\ 442 \end{bmatrix} \bmod 26 = \begin{bmatrix} 19 \\ 12 \end{bmatrix}$

$\begin{bmatrix} 3 & 14 \\ 9 & 15 \end{bmatrix} \begin{bmatrix} 22 \\ 16 \end{bmatrix} \bmod 26 = \begin{bmatrix} 290 \\ 378 \end{bmatrix} \bmod 26 = \begin{bmatrix} 4 \\ 12 \end{bmatrix}$

$$\begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ 12 \end{bmatrix} = \begin{bmatrix} 194 \\ 78 \end{bmatrix} \bmod 26 = \begin{bmatrix} 12 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 6 \end{bmatrix} = \begin{bmatrix} 84 \\ 30 \end{bmatrix} \bmod 26 = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

plain text = TOPSECRETMESSAGE

2. Consider the Hill cipher key matrix $K = \begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix}$

cipher text = RXADDY . Find plain text

Inverse

$$\begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix} = (24 \times 14) - (5 \times 19) = 241 \bmod 26 = \text{Rem } 9 \rightarrow 26 \times 9 = 234 \cdot 241 = 7 \bmod 26$$

$$7^{-1} \bmod 26 = 15$$

$$15 \begin{bmatrix} 14 & -19 \\ -5 & 24 \end{bmatrix} = \begin{bmatrix} 210 & -285 \\ -75 & 360 \end{bmatrix} \bmod 26 = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 17 \\ 23 \end{bmatrix} = \begin{bmatrix} 2 \times 17 + 1 \times 23 \\ 3 \times 17 + 22 \times 23 \end{bmatrix} = \begin{bmatrix} 57 \\ 557 \end{bmatrix} \bmod 26 = \begin{bmatrix} 5 \\ 11 \end{bmatrix} = \begin{bmatrix} F \\ L \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 0 \\ 14 \end{bmatrix} = \begin{bmatrix} 14 \\ 308 \end{bmatrix} \bmod 26 = \begin{bmatrix} 14 \\ 22 \end{bmatrix} = \begin{bmatrix} O \\ W \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 3 \\ 24 \end{bmatrix} = \begin{bmatrix} 30 \\ 537 \end{bmatrix} \bmod 26 = \begin{bmatrix} 4 \\ 17 \end{bmatrix} = \begin{bmatrix} E \\ R \end{bmatrix}$$

plain text = FLOWER

3. Confusion vs Diffusion

Confusion: It hides the relationship between the plaintext & cipher text by changing the value of characters.

Diffusion: It spreads the information of the plaintext by changing the position of characters.

Cipher	changes character value	changes character position	Confusion / Diffusion
Shift cipher	Yes	NO	Confusion
Multiplicative cipher	Yes	NO	Confusion
Affine cipher	Yes	NO	Confusion
Transposition Cipher	NO	Yes	Diffusion

→ Relation ship between substitution & confusion.

- substitution changes one character into another character, Therefore, substitution mainly provides confusion.

→ Relationship between transposition & Diffusion.

- Transposition does not change the character. It only changes their position. Therefore, transposition mainly provides diffusion.

4. Transposition Encryption

Encrypt the message "The arrow is pointing north direction" using

i) key = 4

T H E A
R R O W
I S P O
I N T I
N G N O
R T H D
I R E C
T I O N

cipher text:

TRIINRITHRSNGTRI EOPTNHEOALWDI
ODCN

ii) Key = ATLAS

ATLAS
15324

A T L A S
1 5 3 2 4
T H E A R
R O W I S
P O I N T
N O R T H
D I R E C
T I O N X

cipher text:

TRPNDTAINTENEWIRRD
RSTHCXHOODIX

[considered only "point" not "pointing"]

iii) Double

Key 1 = BREAK

Key 2 = LOG

B R E A K
2 5 4 1 3
T H E A R
R O W I S
P O I N T
I N G N O
R T H D I
R E C T I
O N X X X

cipher text: AINNDTXTRPIR
RORSTOIIXEWIGHCXHOON
TEN

L	O	G
2	3	1
T	H	E
A	R	R
O	W	I
S	P	O
I	N	T
I	N	G
N	O	R
T	H	D
E	R	E
T	F	E
O	N	X

Ciphertext: ERIOTGRDEIXTAOSIINTICO

HRWPN NOHRTN

3. Transposition Decryption.

i) Ciphertext: TIDIE MRNME REOGO SANOW RSLTP EEEDO

SSNV AHIVD BIENP GODAE PEIXD ELNSY = 60/10 = 6

Key = 10

T	R	O	O	P	S	T	N	E	E
I	N	G	W	E	S	V	P	P	L
D	M	U	R	E	N	D	G	E	N
I	E	S	S	E	V	B	O	I	S
E	R	A	L	D	A	I	D	X	Y
M	E	N	T	O	H	E	A	D	X

Plain text = TROOPS HEADING WEST NEED MORE SUPPLIES SEND

GENERAL DUBOIS MEN TO AID

ii) Key = SECRET

Cipher text : ILEIT NHPTR SOIIT MAXSA XERXT ANONI

SNIFOTX = 36/6 = 6

	S	E	C	R	E	T
	5	2	1	4	3	6
T	H	I	S	I	S	
A	P	L	A	I	N	
N	T	E	X	T	F	
O	R	D	E	M	O	
N	S	T	R	A	T	
I	D	N	X	X	X	

Plain text:

THIS IS A PLAIN TEXT FOR DEMOSTRATION

iii) Key : PLAIN ciphertext: VEY EAX ARA ATX HGO = 15/5 = 3

	P	L	A	I	N
	5	3	1	2	4
H	A	V	E	A	
G	R	E	A	T	
D	A	Y	X	X	

Plain text:

HAVE A GREAT DAY

6. permutation, Find the inverse permutation for.

i) 4 3 1 5 2

original 1 2 3 4 5

position 4 3 1 5 2

P^{-1} = 3 5 2 1 4

ii) 7 4 1 3 5 2 6

P → 1 2 3 4 5 6 7

7 4 1 3 5 2 6

P^{-1} = 3 6 4 2 5 7 1

iii) 3 5 1 4 2

P → 1 2 3 4 5

3 5 1 4 2

P^{-1} = 3 5 1 4 2

iv) 4 1 3 2

P → 1 2 3 4

4 1 3 2

P^{-1} = 2 4 3 1